

# MohammadArman Maghsoudi

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## Education

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| <b>Politecnico di Milano</b><br>M.Sc. in Management Engineering (Track: Analytics for Business)    | <i>Sep 2025 – Jul 2027</i> |
| <b>Sharif University of Technology</b><br>B.Sc. in Industrial Engineering<br>Final Grade: 18.01/20 | <i>Sep 2020 – May 2025</i> |

## Experience

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| <b>Research Assistant</b> — Sharif University of Technology                                                                                                                                                                                                                                                                   | <i>Sep 2024 – Present</i>  |
| <ul style="list-style-type: none"><li>Conducted extended research on Operating Room Scheduling to develop a publishable paper based on my bachelor thesis.</li><li>Collaborated with faculty advisors on literature review, model validation, and manuscript preparation.</li></ul>                                           |                            |
| <b>Quality Control &amp; Operations Intern</b> — Shahid Bahonar Copper Industries Co. (CSP)                                                                                                                                                                                                                                   | <i>Jul 2024 – Sep 2024</i> |
| <ul style="list-style-type: none"><li>Improved production efficiency through analysis of quality metrics and Industrial Engineering methodologies.</li><li>Collaborated with cross-functional teams to ensure compliance with industrial standards.</li></ul>                                                                 |                            |
| <b>Teaching Assistant</b> — Sharif University of Technology                                                                                                                                                                                                                                                                   | <i>Feb 2023 – Mar 2025</i> |
| <ul style="list-style-type: none"><li>Courses: Operations Research II, Queuing Theory, Project Control, Quality Control, Engineering Economics</li><li>Led problem-solving sessions in optimization, stochastic modeling, and computational methods, and designed and graded undergraduate assignments and quizzes.</li></ul> |                            |

## Related Projects & Research

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| <b>Research Interests:</b> Optimization under uncertainty (stochastic, simulation-based), Combinatorial optimization (MIP, heuristics, metaheuristics), Queueing systems, Data-driven decision-making.                   |             |
| <b>Operating Room Scheduling — Genetic Algorithm &amp; MIP (Bachelor Thesis)</b>                                                                                                                                         | <i>2025</i> |
| <ul style="list-style-type: none"><li>Developed GA and mixed-integer optimization models in Python using PuLP.</li><li>Performed a scenario-based simulation to account for uncertainty in surgical schedules.</li></ul> |             |
| <b>Hospital Patient Flow Optimization — Discrete-Event Simulation (Principles of Simulation Course)</b>                                                                                                                  | <i>2025</i> |
| <ul style="list-style-type: none"><li>Built simulation models using queueing theory and stochastic processes.</li><li>Evaluated resource allocation policies to improve system stability.</li></ul>                      |             |
| <b>Bike-Sharing Demand Forecasting &amp; Facility Location Optimization (Transportation Planning Course)</b>                                                                                                             | <i>2024</i> |
| <ul style="list-style-type: none"><li>Applied Machine Learning forecasting models and OR-based location-allocation optimization.</li></ul>                                                                               |             |
| <b>Agricultural &amp; Industrial Planning — Linear Optimization (Operations Research II Course)</b>                                                                                                                      | <i>2023</i> |
| <ul style="list-style-type: none"><li>Developed LP models using CPLEX, GAMS, and Python; performed sensitivity and duality analysis.</li></ul>                                                                           |             |

## Manuscripts

Maghsoudi, M.A., Radman, M, Development of an Elite-Oriented Genetic Algorithm to Solve the Operating Room Scheduling Problem Under Uncertainty of Surgical Time.Submitted to Sharif Journal of Industrial Engineering & Management, under review (Nov 2025).

## Technical Skills

**Optimization Tools:** Pyomo, Gurobi, Pulp, CPLEX, GAMS, LINGO  
**Programming Tools:** Python(Pandas, NumPy, SciPy, SciKit-learn), SQL, Java,  $\text{\LaTeX}$   
**Software:** Power BI, Tableau, Minitab, MS Office, GitLab, Vensim, Visual Paradigm, Canva, Figma

## Awards & Honors

Nominated for IAESTE International Internships (in Vietnam & Nepal) to contribute to Machine Learning projects. · 2024  
Top 0.6% in the National University Entrance Exam of Iran (Math & Physics) · 2020  
Full Scholarship — Sharif University of Technology · 2020  
Finalist — Iranian Geometry Olympiad & Tournament of Towns · 2016 - 2018

## Languages

English (TOEFL 112) · German (B1) · Italian (Elementary) · Persian (Native)

## Certificates

Game Theory — Stanford University (MOOC) · 2025  
Analyzing & Visualizing Data with Power BI — University of Tehran · 2023

## Hobbies

Reading, Traveling, Track & field, Football, Tennis